PHYS3114: Electrodynamics Assignment

1. Green's Function with mixed boundary Conditions

On the interval $0 < x < L$, the electrostatic potential satisfies,

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = -\frac{\rho_0}{\varepsilon_0}, \quad \varphi(0) = 0, \quad \varphi'(L) = 0, \tag{1}$$

where ρ_0 is constant.

a) Constructing a Green's Function.

Given the following conditions in which the Green Function satisfies,

$$\frac{\partial^2 G(x,x')}{\partial x^2} = -\frac{\delta(x-x')}{c_0}, \quad G(0,x') = 0, \quad \left.\frac{\partial G}{\partial x}\right|_{x=L} = 0 \tag{1}$$

Using the fact we are given the potential at one side of the boundary ($x = 0$), and the derivative of the normal to on the other end of the potential ($x = L$). We have to use both descriptions to ascribe a Green's Function that satisfies the mixed boundary conditions. First consider when $x \neq x'$, the first term in Equation 1 becomes,

$$\frac{\mathrm{d}^2 G(x,x')}{\mathrm{d}x^2} = 0 \tag{2}$$

This implies the Green's Function is linear in nature, therefore we can write it as,

$$G(x,x') = \begin{cases} Ax & x < x' \\ B + C(L-x) & x' < x < L. \end{cases} \tag{3}$$

Consider the fact the first derivative of the Green's function has been given to us on the boundary,

$$\left.\frac{\partial G}{\partial x}\right|_{x=L} = CL = 0, \tag{4}$$

therefore, this defines that $C = 0$. Rewriting our Green's function with this gives us,

$$G(x,x') = \begin{cases} Ax & x < x' \\ B & x' < x < L. \end{cases} \tag{5}$$

We need that the Green's Function at $x = x'$ be continuous to be at the discontinuity to satisfy the delta function. We can write this out as,

$$Ax = B \quad \text{as } x \to x' \tag{6}$$

Taking the partial with respect to x' from both sides gives us the following relationship the function must satisfy,

$$\begin{aligned} -\left.\frac{\partial G}{\partial x}\right|_{x\to x'_-} + \left.\frac{\partial G}{\partial x}\right|_{x\to x'_+} &= \frac{\partial G}{\partial x} \\ -A + 0 &= \frac{1}{\varepsilon_0} \\ A &= \frac{1}{\varepsilon_0} \end{aligned} \tag{7}$$

Substituting this into Equation 6,

$$B = \frac{x'}{\varepsilon_0} \tag{8}$$

Therefore, we can define the final Green Function as,

$$G(x, x') = \begin{cases} \frac{x}{\varepsilon_0} & x < x' \\ \frac{x'}{\varepsilon_0} & x' < x < L. \end{cases} \tag{9}$$

□

b) Verify the derived Green's Function is symmetric under $x \leftrightarrow x'$

Let's consider the case where initially $x < x'$, such that

$$G(x, x') = \frac{x}{\varepsilon_0}, \tag{1}$$

we then perform the transformation. We can see by inspection that the function satisfies the other case so that

$$G(x, x') = \frac{x'}{\varepsilon_0} \tag{2}$$

but the transformation turns the $x \to x'$, so the solution becomes,

$$G(x', x) = \frac{x}{\varepsilon_0}. \tag{3}$$

By equating Equation 1 and Equation 3, we get that the derived Green's function is symmetric,

$$G(x, x') = G(x', x) \tag{4}$$

We can use a similar derivation to prove for the other initial case.

□

c) Show that the boundary terms in Green's identity vanish for the stated boundary conditions.

Using Green's second identity for the general three dimensional case,

$$\int_V \mathrm{d}^3\boldsymbol{r}[\varphi\nabla^2 G - G\nabla^2\varphi] = \oint_S \mathrm{d}\boldsymbol{S} \cdot (\varphi\nabla G - G\nabla\varphi) \tag{1}$$

As our problem is restricted along 1 dimension, we can rewrite the identity in a more useful manner,

$$\int_0^L \left(\varphi\frac{\partial^2 G}{\partial x'^2} - G\frac{\mathrm{d}^2\varphi}{\mathrm{d}x'^2}\right)\mathrm{d}x' = \left[\varphi\frac{\partial G}{\partial x'} - G\frac{\mathrm{d}\varphi}{\mathrm{d}x'}\right]_0^L \tag{2}$$

Due to the symmetry of the solution as proved above, the Green's Function boundary conditions can be ascribed to both x and x'. Using the boundary conditions, we can rewrite the Equation 2 as

$$\int_0^L \varphi\left(-\frac{\delta(x-x')}{\varepsilon_0}\right)\mathrm{d}x' - \int_0^L G\left(-\frac{\rho_0}{\varepsilon_0}\right)\mathrm{d}x' = \varphi\frac{\partial G}{\partial x'} - G\frac{\mathrm{d}\varphi}{\mathrm{d}x'}\bigg|_{x'=L} - \left(\varphi\frac{\partial G}{\partial x'} - G\frac{\mathrm{d}\varphi}{\mathrm{d}x'}\bigg|_{x'=0}\right) \tag{3}$$

Note that the RHS evaluated at their limits all cancel due to the boundary conditions, thus leaving us with only the LHS,

$$-\frac{1}{\varepsilon_0}\int_0^L \varphi(x')\delta(x-x')\,\mathrm{d}x' = -\frac{1}{\varepsilon_0}\int_0^L G(x, x')\rho_0\,\mathrm{d}x' \tag{4}$$

The LHS integral is evaluated with respect to the delta function at x', this can be rearranged into the required expression,

$$\begin{aligned} -\frac{1}{\varepsilon_0}\varphi(x) &= -\frac{1}{\varepsilon_0}\int_0^L \rho_0 G(x,x')\,\mathrm{d}x' \\ \varphi(x) &= \int_0^L \rho_0 G(x,x')\,\mathrm{d}x' \end{aligned} \tag{5}$$

As required.

□

d) Evaluate the integral and obtain $\varphi(x)$ and $E_{x(x)}$. Checking the boundary conditions and Poisson equation.

Consider the integral to find the potential of this system,

$$\varphi(x) = \int_0^L \rho_0 G(x,x')\,\mathrm{d}x' \tag{1}$$

We know the Green's function can be described by,

$$G(x,x') = \begin{cases} \frac{x}{\varepsilon_0} & x < x' \\ \frac{x'}{\varepsilon_0} & x' < x < L. \end{cases} \tag{2}$$

Substituting this into the integral gives us,

$$\begin{aligned} \varphi(x) &= \int_0^x \frac{x'}{\varepsilon_0}\rho_0\,\mathrm{d}x' + \int_x^L \frac{x}{\varepsilon_0}\rho_0\,\mathrm{d}x', \\ \varphi(x) &= \frac{\rho_0}{\varepsilon_0}\left[\frac{x'^2}{2}\right]_0^x + \frac{\rho_0 x}{\varepsilon_0}[x']_x^L, \\ &= \frac{\rho_0}{\varepsilon_0}\left(\frac{x^2}{2} + x(L-x)\right). \end{aligned} \tag{3}$$

Simplifying this result gives us the potential in x as,

$$\varphi(x) = \frac{\rho_0}{\varepsilon_0}\frac{x}{2}(2L-x). \tag{4}$$

The electric field in the x-direction can be found by taking the derivative with respect to x,

$$E_x(x) = -\frac{\mathrm{d}\varphi}{\mathrm{d}x} = -\frac{\rho_0}{\varepsilon_0}(L-x). \tag{5}$$

Checking our boundary conditions to double check our solution,

$$\begin{aligned} \varphi(0) &= \frac{\rho_0}{\varepsilon_0}\frac{0}{2}(2L-0) = 0 \\ \varphi'(L) &= -E_x(L) = -\frac{\rho_0}{\varepsilon_0}(L-L) = 0 \end{aligned} \tag{6}$$

By inspection the potential satisfies both the Neumann and Dirichlet conditions defined on the boundary as required. Similarly, we can check the Poisson equation,

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = -\frac{\mathrm{d}E_x}{\mathrm{d}x} = -\frac{\rho_0}{\varepsilon_0} \tag{7}$$

Therefore, also satisfying the Poisson equation.

□

e) Find the potential at the boundary L

By substitution of $x = L$, we get the potential at the boundary is,

$$\varphi(L) = \frac{\rho_0}{\varepsilon_0}\frac{L^2}{2} \tag{1}$$

as required.